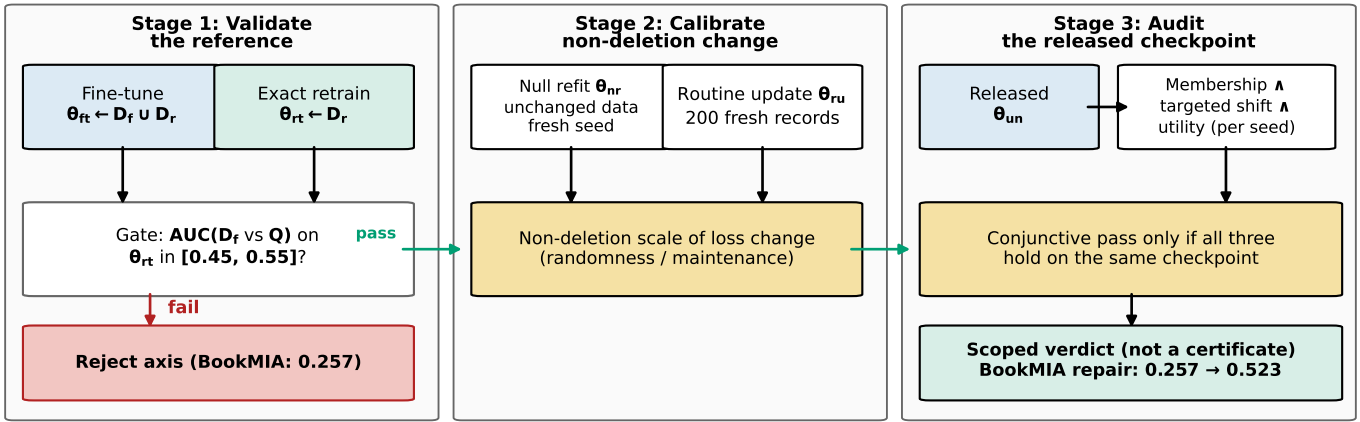
Stage 1: Validate the reference
Fine-tune $\theta_{ft} \leftarrow D_f \cup D_r$
Exact retrain $\theta_{rt} \leftarrow D_r$
Gate: AUC(D_f vs Q) on θ_{rt} in [0.45, 0.55]?
pass
fail
Reject axis (BookMIA: 0.257)
Stage 2: Calibrate non-deletion change
Null refit θ_{nr} unchanged data fresh seed
Routine update θ_{ru} 200 fresh records
Non-deletion scale of loss change (randomness / maintenance)
Stage 3: Audit the released checkpoint
Released θ_{un}
Membership ∧ targeted shift ∧ utility (per seed)
Conjunctive pass only if all three hold on the same checkpoint
Scoped verdict (not a certificate)
BookMIA repair: 0.257 → 0.523